

## EMPLOYMENT

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### LexisNexis Legal & Professional

Raleigh, NC

Senior Data Scientist I

May 2024 - Present

- Technical lead for an agentic legal drafting system; led its migration from RAG and continue to own development.
- Reduced customer-facing product runtime by 24.8% through LLM call profiling and parallelization.
- Profiled high pod memory usage and introduced lazy module loading, reducing time at peak memory from 86.4% to 68% of runtime.
- On a 3-person team, increased the share of drafts rated suitable for users by attorney reviewers from 40% to 86% for complaints and 59.2% to 76% for motions.
- Built a contrastive-learning filter to remove unrelated legal passages from search results; median filtering latency of 12ms across 40 queries.
- Mentored an intern on a large-scale, full factorial experiment investigating Pareto-optimal configurations for draft quality and latency.
- Built team developer tooling: a Rust MITM proxy with request replay, a web UI for visualizing intermediate agent steps, and custom Python lint rules beyond `ruff`'s rule set.
- Technology:** Python, Litestar, Google ADK, React, Rust, TypeScript, Tailwind

### Amazon

New York, NY / Bellevue, WA

Software Dev Engineer Intern

May 2023 - Aug 2023

- Implemented and shipped profile locks for Prime Video on Echo Show devices.
- Technology:** React Native, TypeScript

Software Dev Engineer Intern

May 2022 - Jul 2022

- Developed a full-stack system to publish announcements in scorecards used by delivery service partners.
- Technology:** React/Redux, TypeScript, Redux Saga, DynamoDB, Java Spring

## RELEVANT PROJECTS

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### pi-permission-system (maintained fork)

May 2026 - Present

TypeScript, Node.js

[GitHub](#)

Maintain a permission-system fork for the Pi coding agent. Redesigned shell permissions using grammar-based analysis of pipelines, substitutions, and redirections, with protected-path checks and argument-aware approval rules. Extended enforcement to background commands and fixed concurrent subagent approval handling.

### Q&A System for Zotero

Oct 2024 - Present

Rust, SQLite, LanceDB

[GitHub](#) :: [crates.io](#)

Built an academic question-answering and semantic search system over a user's Zotero library. Wrote a custom PDF parser for academic papers (median throughput  $\approx$  2,400 pages/s across 11 PDFs; M4 Pro) and a RAG library with support for multiple providers and tool calling.

### JournalBear

Jan 2017 - Jul 2026

TypeScript, React, Zustand, Electron, SwiftUI

[GitHub](#) :: [App Store](#) :: [Softpedia](#)

Cross-platform journaling app with AES-256 encryption and separate native apps for iOS and macOS. Rated 4/5 by Softpedia.

### The 1 Billion Row Challenge (1BRC)

Feb 2026 - Apr 2026

Zig

[GitHub](#)

Wrote a Zig implementation for the 1BRC, running in 1.20s (M4 Pro, 14 core) using mmap, open addressing (Wyhash + Robin Hood probing), and packed SIMD.

### Programmable Resumes

Aug 2023 - Mar 2025

Python

[GitHub](#)

Developed a specification and implementation for modular, customizable resumes with support for two popular LaTeX templates. Wrote this resume using this tool.

SKILLS

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**Languages:** Rust, Python, TypeScript, Java, C++, Gleam  
**ML/AI:** Keras, PyTorch, Hyper-parameter optimization, RAG, LLM evaluation, Agentic systems, Google ADK  
**Frameworks:** React, Flask, Litestar, FastAPI  
**Databases:** LanceDB, MySQL, MongoDB, DynamoDB  
**Cloud:** Google Cloud (Compute Engine, Cloud Storage, Vertex AI), AWS (EC2, S3, DynamoDB)

EDUCATION

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| <b>North Carolina State University</b><br><i>Ph.D. Computer Science</i> <ul style="list-style-type: none"><li>◦ Advisor: <i>Dr. Tim Menzies</i></li><li>◦ Dissertation: <i>Guidelines for the Application of Neural Technologies in Software Analytics (or: How to Do More with Less in SE)</i></li></ul> | Raleigh, NC<br><i>Aug 2019 - Jul 2024</i>      |
| <b>PES University</b><br><i>B.E. Computer Science – Advisor: Dr. Snehanshu Saha</i>                                                                                                                                                                                                                       | Bangalore, India<br><i>Aug 2015 - May 2019</i> |

HONORS

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|------------------------------------------------------|---------------------|
| <b>Distinguished Reviewer, ICSE 2026</b>             | Mar 2026            |
| <b>Google Developer Expert - ML and GCP</b>          | Mar 2025 - Present  |
| <b>Google Cloud Champion Innovator - Cloud AI/ML</b> | Oct 2022 - Mar 2025 |
| <b>Google Cloud Research Innovator</b>               | Feb 2022            |